

STANDING ORDER: Pine Script ↔ Backend Version Sync Protocol

Effective immediately. **Mandatory** on every Pine Script version increment.

Reason: On 2026-05-17 the production Railway dashboard was caught serving **v17.15** while the Pine Script was at **v17.25**, and the on-page alert templates contained wrong `{{plot_X}}` mappings (`h4_bias=plot_3` instead of `plot_10`, etc.) which would have produced wrong opportunity records for every alert pasted from the dashboard.

The Rule

You may not ship a Pine Script version bump as “complete” unless the backend, templates, frontend, schema, and alert template documentation are updated in the same PR (or an immediately-following sibling PR merged together).

A Pine Script bump on its own is a regression. The dashboard becomes a lie. Alert templates drift silently.

Mandatory Checklist on Every `vX.Y → vX.Z` Increment

Run this checklist **every single time** the Pine Script's `indicator(...)` title changes. No exceptions.

1. Backup

- [] `cp smc_premium_discount_indicator_v<OLD>.pine smc_premium_discount_indicator_v<OLD>_backup.pine`
- [] Snapshot the backend: `cp -r app.py templates src schemas tradex-frontend/app backups/v<OLD>_backend_backup/`

2. Pine Script

- [] Update `indicator(...)` title to new version.
- [] Update changelog header.
- [] Update HUD header cell text.
- [] Update **all four** `alertcondition()` JSON payloads: `sniper_long`, `sniper_short`, `retrace_long`, `retrace_short` — both the `"version": "..."` field and any inline comments.
- [] Confirm plot budget ≤ 64 (`grep -c "^plot\|plotshape\|alertcondition\|bgcolor"`).

3. Backend Version Strings

Update **every** occurrence in:

- [] `app.py` (logger banner)
- [] `src/webhook_server/routes.py` (sample payloads + format-detection comments)
- [] `src/database.py` (migration log messages)
- [] `src/decoders.py` (module header)
- [] `src/oie_database.py` (default version on insert)
- [] `src/oie_processor.py` (header + the **current-version branch only** — keep legacy version)

branches intact for backward compatibility)

- [] `schemas/migrate_v17_*.sql` (header comment + `DEFAULT 'vX.Z'`)

4. HTML Templates

- [] `templates/base.html` — `<title>` + footer badge
- [] `templates/settings.html` — title, section headers, server-info badge, decode-reference label, API-endpoints heading
- [] `templates/dashboard.html` — `<title>`
- [] `templates/trades.html` — `<title>`
- [] `templates/opportunities.html` — `<title>` + page badge

5. Next.js Frontend

- [] `tradex-frontend/app/layout.tsx` — title metadata, header & footer badges
- [] `tradex-frontend/app/page.tsx` — hero subtitle, version badge, decode-reference header
- [] `tradex-frontend/app/opportunities/page.tsx` — subtitle + empty-state copy
- [] `tradex-frontend/app/settings/page.tsx` — section header + all alert JSON blocks
- [] `tradex-frontend/lib/api.ts` — module banner

6. Alert Template Audit (CRITICAL)

Open the Pine Script and read lines containing `alertcondition(...)` . For each of the 4 alerts, write down the **exact** `{{plot_N}}` placement for every JSON field. Then open:

- [] `templates/settings.html` — the four `<pre>` blocks under “Alert Templates”
- [] `tradex-frontend/app/settings/page.tsx` — the four `<pre>` blocks under “Alert Templates”

For **each** of the 4 alerts (`sniper_long` , `sniper_short` , `retrace_long` , `retrace_short`), confirm field-by-field that every `{{plot_N}}` in the docs matches the Pine Script. **Never copy from the previous version’s template** — always re-derive from the Pine Script’s `alertcondition()` line for the new version.

Currently (v17.25) the canonical map is:

<code>plot_0</code> → <code>entryLong</code>	<code>plot_7</code> → <code>poiScore</code>
<code>plot_1</code> → <code>slLong</code>	<code>plot_8</code> → <code>confluenceScore</code>
<code>plot_2</code> → <code>tpLong</code>	<code>plot_9</code> → <code>dtStage</code>
<code>plot_3</code> → <code>entryShort</code>	<code>plot_10</code> → <code>h4Bias</code>
<code>plot_4</code> → <code>slShort</code>	<code>plot_11</code> → <code>pdZone</code>
<code>plot_5</code> → <code>tpShort</code>	<code>plot_12</code> → <code>kzCode (kill_zone)</code>
<code>plot_6</code> → <code>qualityScore</code>	<code>plot_13</code> → <code>guardianCode</code>

If the Pine Script rennumbers plots in a future version, this map **must** be re-derived and the templates **must** be regenerated.

7. Validation

- [] `python3 -m py_compile app.py src/**/*.py` → exit 0
- [] Jinja2 parse: `for t in templates/*.html; do python3 -c "from jinja2 import Environment, FileSystemLoader; Environment(loader=FileSystemLoader('templates')).get_template('$ (basename $t)')"; done`
- [] Flask test client renders every page with the new version visible and no stale version strings.
- [] `cd tradex-frontend && npx tsc --noEmit` → exit 0.

- [] `grep -rEn "v<OLD>" --include="*.py" --include="*.html" --include="*.tsx" --include="*.ts" --include="*.sql" | grep -v "backups/\|test_oie_integration\|smc_premium_discount_indicator" → empty.`

8. Deploy Verification

- [] After merging to `main`, wait for Railway redeploy.
- [] `curl -s https://web-production-b63af.up.railway.app/settings | grep -c "v<NEW>" → ≥ 5.`
- [] After merging, wait for Vercel redeploy.
- [] `curl -s https://smc-tracker-railway-2027.vercel.app/settings | grep -c "v<NEW>" → ≥ 5.`

9. Reporting

- [] Write `v<NEW>_completion_report.md` covering Pine Script changes.
- [] Write `v<NEW>_backend_sync_report.md` covering backend + frontend + template audit.

Anti-patterns (Do Not Do These)

-  Shipping a Pine Script bump on TradingView before the backend is updated.
 -  Updating only the `<title>` and `<badge>` and skipping the alert template audit.
 -  Copying the previous version's settings.html alert blocks forward without re-checking the Pine Script's `alertcondition()` plot indices.
 -  Touching `oie_processor.py`'s `v17.14` / `v17.12.3` compatibility branches when sweeping version strings — those are intentional legacy handlers.
 -  Modifying `test_oie_integration.py`'s legacy payloads — they're there to prove the backend still accepts old alerts.
 -  Force-pushing or skipping the per-version backup directory.
-

When You're Done

You should be able to answer **YES** to every one of these:

1. Does `https://web-production-b63af.up.railway.app/settings` display the new version?
2. Does `https://smc-tracker-railway-2027.vercel.app/settings` display the new version?
3. If a user copies a Sniper Long template from `/settings` into TradingView **right now**, will the decoded opportunity row show correct `h4_bias`, `p_d_zone`, `guardian`, `quality`, and `poi` values?
4. Is there a backup of the previous backend state under `backups/v<OLD>_backend_backup/` ?
5. Are the completion report **and** the backend-sync report both present in the repo root?

If any answer is "no", **the release is not complete**.

Last enforced: v17.25 (2026-05-17) — see `v17.25_backend_sync_report.md`.